



Presented by Aryan Pattnaik



Atliq Hardwares

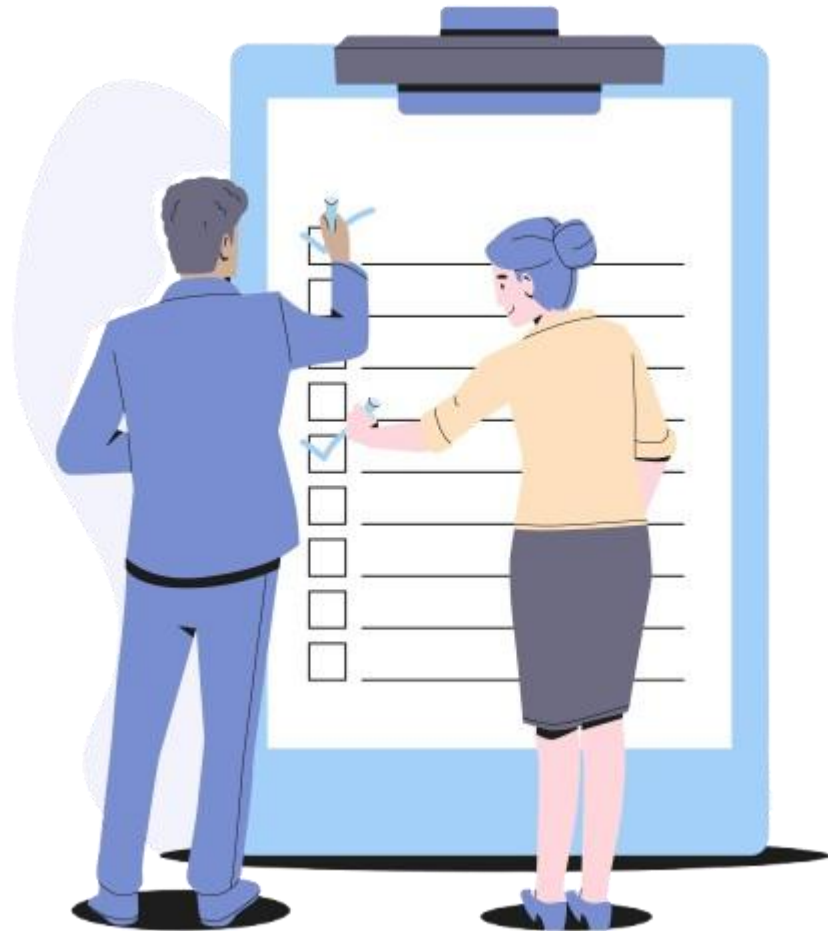
CONSUMER GOODS AD-HOC INSIGHTS

SQL Project Challenge

AGENDA

1. Background/Context
2. Getting familiar with Atliq's Business - Their Markets and Product lines
3. Getting familiar with the input data
4. Ad-hoc requests along with the queried result, visualizations and insights





BACKGROUND/CONTEXT

Our Company - Atliq Hardwares (imaginary company) - One of the leading computer hardware producers in India.

Background- The management noticed that they do not get enough insights to make quick and smart data-informed decisions.

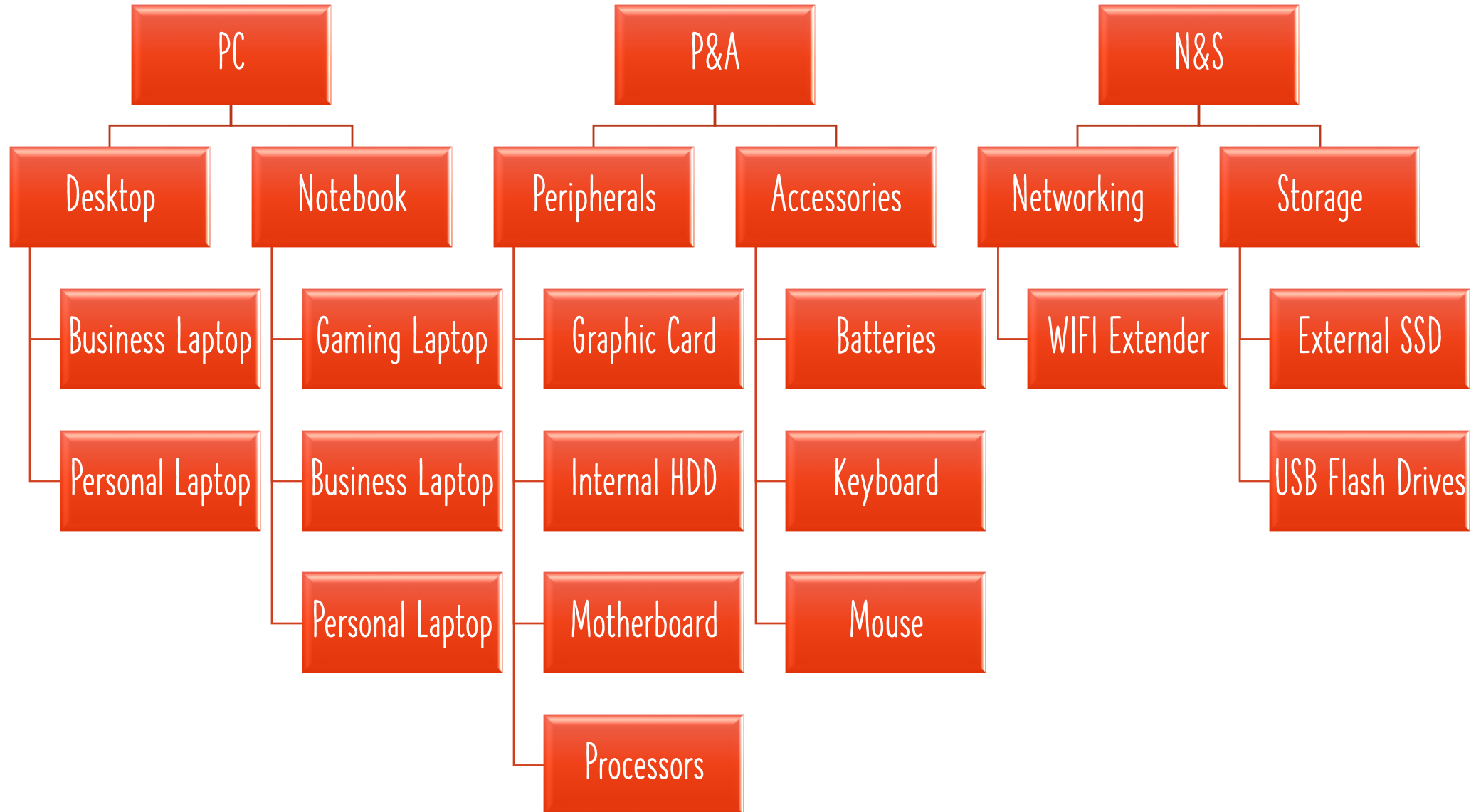
Problem- There are 10 ad-hoc requests for which the company needs insights.

Approach - Run a SQL query to answer these requests. Convert it into visualizations and present the Insights to the toplevel management.

ATLIQ'S MARKETS



PRODUCT LINES



Input data consists of sales data for FY 2020 and FY 2021, along with different other dimension tables like customer details, product details, etc.



dim_customer
customer_code INT
customer VARCHAR(150)
platform VARCHAR(45)
channel VARCHAR(45)
market VARCHAR(45)
sub_zone VARCHAR(45)
region VARCHAR(45)

fact_pre_invoice_deductions
customer_code INT
fiscal_year YEAR
pre_invoice_discount_pct DECIMAL(5,4)

fact_gross_price
product_code VARCHAR(45)
fiscal_year YEAR
gross_price DECIMAL(15,4)

fact_sales_monthly
date DATE
product_code VARCHAR(45)
customer_code INT
sold_quantity INT
fiscal_year YEAR

fact_manufacturing_cost
product_code VARCHAR(45)
cost_year YEAR
manufacturing_cost DECIMAL(15,4)

dim_product
product_code VARCHAR(45)
division VARCHAR(45)
segment VARCHAR(45)
category VARCHAR(45)
product VARCHAR(200)
variant VARCHAR(45)

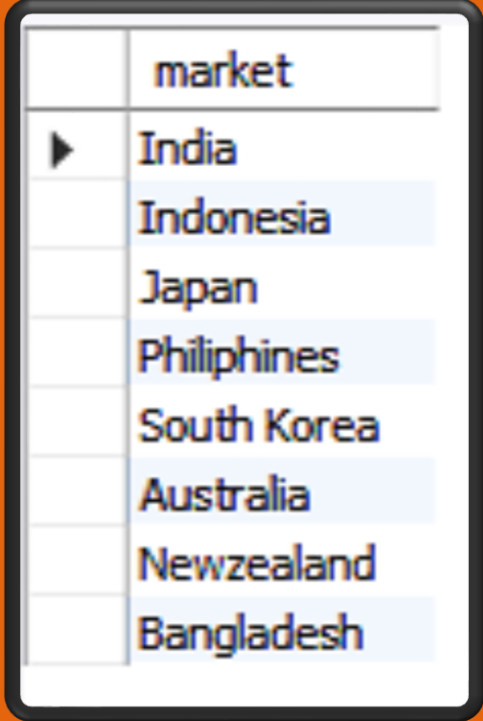
Ad-hoc requests
Queried Results
Insights
And
Visualization



Request 1 :-

Provide the list of markets in which customer "Atliq Exclusive" operates its business in the APAC region.

Output :-

A screenshot of a web application interface. It shows a dropdown menu with a list of countries. The first item, 'India', is highlighted with a blue background and a small black triangle icon to its left. The other items are 'Indonesia', 'Japan', 'Philippines', 'South Korea', 'Australia', 'Newzealand', and 'Bangladesh'. The dropdown menu is enclosed in a rounded rectangle with a dark border.

	market
▶	India
	Indonesia
	Japan
	Philippines
	South Korea
	Australia
	Newzealand
	Bangladesh

Insights :-

This analysis identifies all APAC markets where the customer *Atliq Exclusive* operates. It helps understand geographic presence and regional coverage of the brand within the APAC business unit. Our Exclusive store has established its presence in 8 major markets.



Request 2 :-

What is the percentage of unique product increase in 2021 vs. 2020?

The final output contains these fields

- unique_products_2020
- unique_products_2021
- percentage_chg

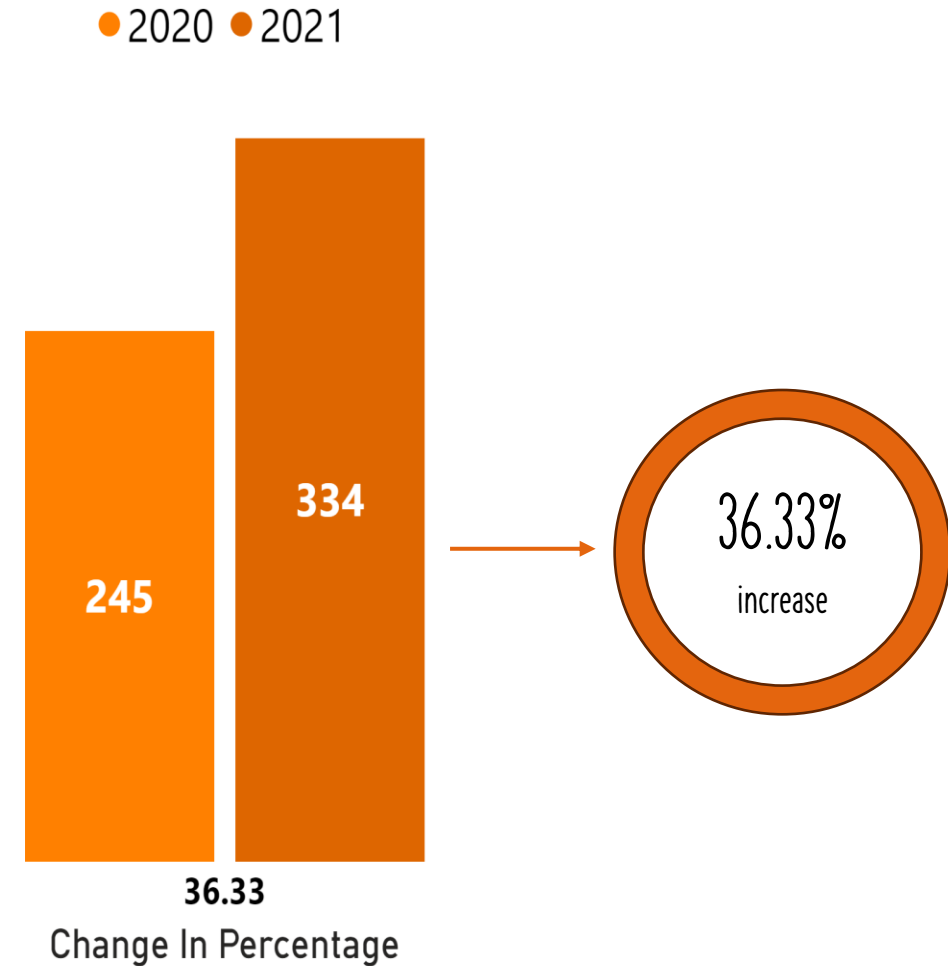
Output :-

	unique_products_2020	unique_products_2021	percentage_chg
▶	245	334	36.33

Insights :-

This analysis measures product portfolio expansion by comparing the number of unique products launched in 2021 versus 2020. The percentage change indicates growth in product diversification strategy.

It's a good sign that we are continuously innovating and introducing new products to the market. In FY 2020, we had a total of 245 products, but in FY 2021, our count increased by 36% to 334 products.



Request 3 :-

Provide a report with all the unique product counts for each segment and sort them in descending order of product counts.

The final output contains 2 fields

- segment
- product_count

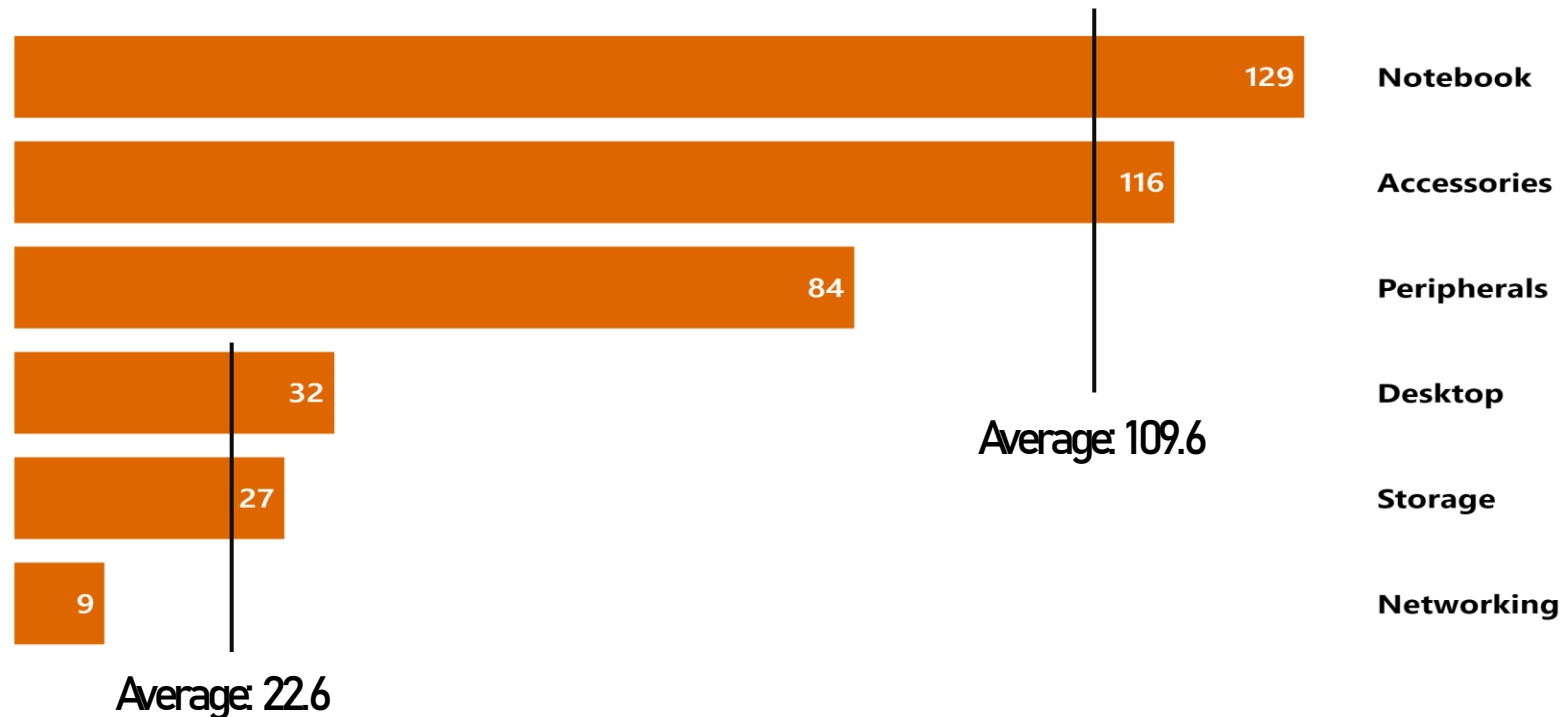
Output :-

	segment	product_count
▶	Notebook	129
	Accessories	116
	Peripherals	84
	Desktop	32
	Storage	27
	Networking	9

Insights :-

This report shows the distribution of products across different segments. It highlights which segments contribute most to the overall product portfolio.

We provide a wide range of products under the segments Notebook, Peripherals, and Accessories, with an average of 110 products in each segment. However we still need to diversify our production in the Desktop, Networking, and Storage segments, where there are just an average of only 23 products per segment.



Request 4 :-

Follow-up: Which segment had the most increase in unique products in 2021 vs 2020?

The final output contains these fields

- segment product_count_2020
- product_count_2021
- difference

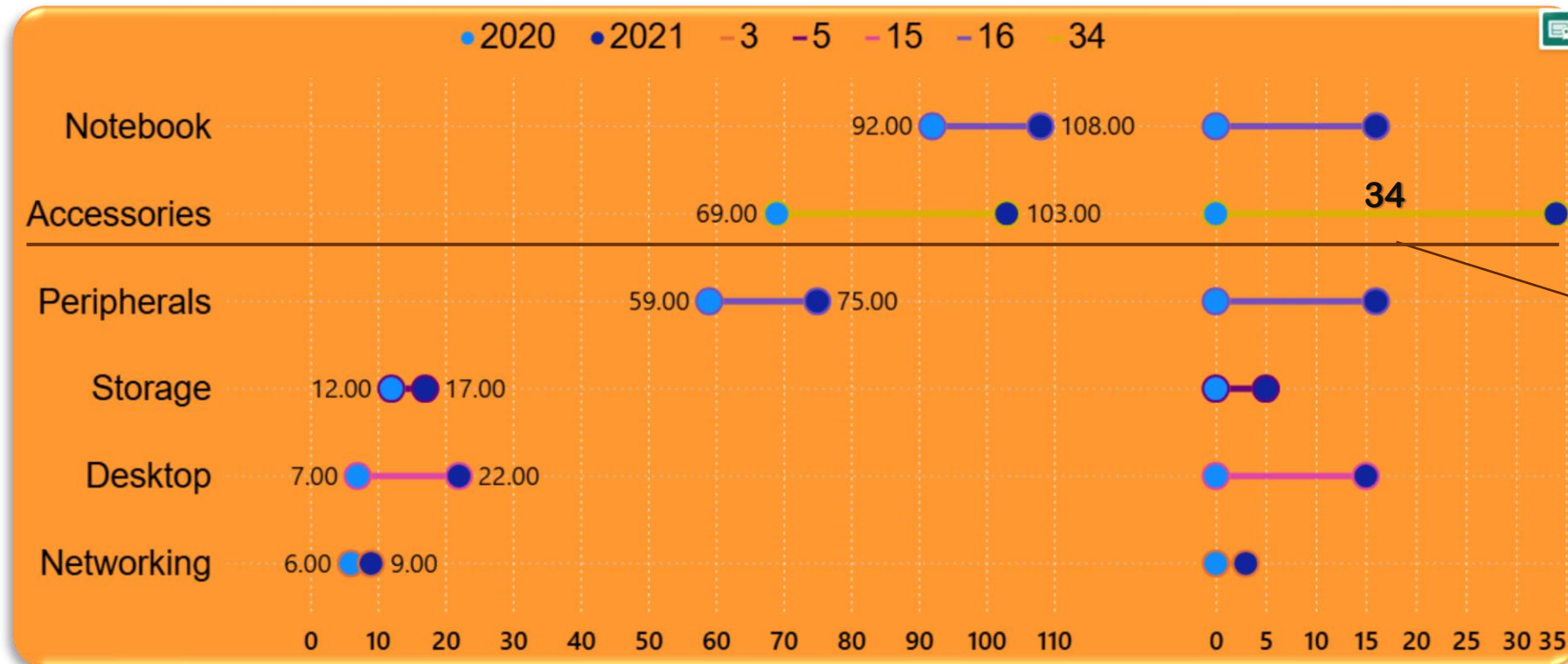
Output :-

	segment	product_count_2020	product_count_2021	difference
▶	Accessories	69	103	34
	Notebook	92	108	16
	Peripherals	59	75	16
	Desktop	7	22	15
	Storage	12	17	5
	Networking	6	9	3

Insights :-

This comparison identifies which product segment experienced the highest increase in unique products between 2020 and 2021, indicating strategic expansion focus areas.

In 2021, we were mainly focusing on diversifying our accessories segment. We introduced 34 new products to the market in accessories.



This segment had the most increase

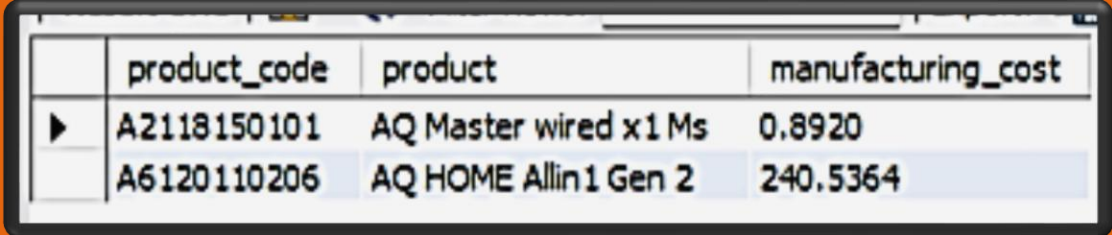
Request 5 :-

Get the products that have the highest and lowest manufacturing costs.

The final output should contain these fields

- product_code
- product
- manufacturing_cost

Output :-

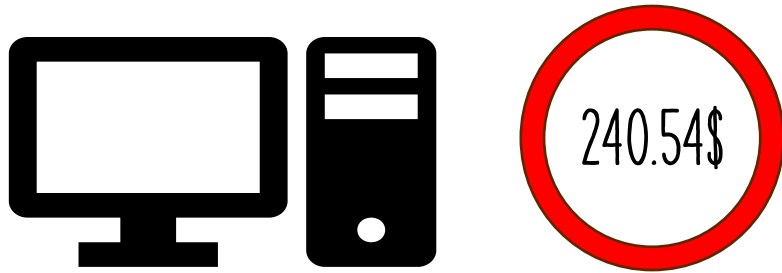


	product_code	product	manufacturing_cost
▶	A2118150101	AQ Master wired x1 Ms	0.8920
	A6120110206	AQ HOME Allin1 Gen 2	240.5364

Insights :-

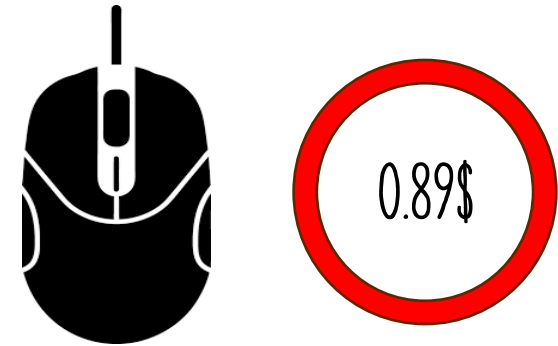
This analysis identifies products with the highest and lowest manufacturing costs. It helps evaluate cost efficiency and potential margin impact across the product portfolio.

The Highest Manufacturing Cost



Personal Desktop
AQ HOME Allin1 Gen 2 (Plus 3)

The Lowest Manufacturing Cost



Mouse
AQ Master wired x1 Ms (Standard 1)

Request 6 :-

Generate a report which contains the top 5 customers who received an average high pre_invoice_discount_pct for the fiscal year 2021 and in the Indian market.

The final output contains these fields

- customer_code
- customer
- average_discount_percentage

Output :-

	customer_code	customer	average_discount_percentage
▶	90002009	Flipkart	0.3083
	90002006	Viveks	0.3038
	90002003	Ezone	0.3028
	90002002	Croma	0.3025
	90002016	Amazon	0.2933

Insights :-

This report highlights the top 5 customers in India who received the highest average pre-invoice discount in FY2021. It helps assess discount concentration and pricing strategy effectiveness.

In 2021, we offered nearly equal pre-invoice discount percentages to each of our top 5 customers, given that Flipkart is the most discounted customer in the Indian market, which equals 30.83%.



Amazon

29.33%

Request 7 :-

Get the complete report of the Gross sales amount for the customer "Atliq Exclusive" for each month . This analysis helps to get an idea of low and high-performing months and take strategic decisions.

The final report contains these columns:

- Month
- Year
- Gross sales Amount

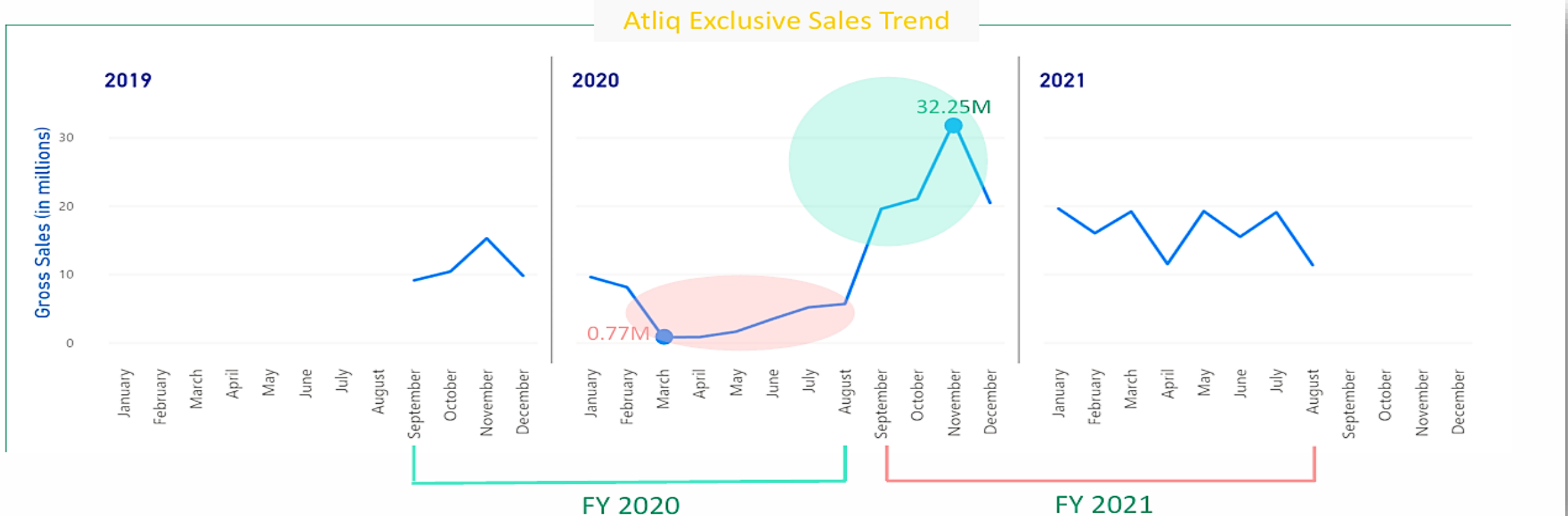
Output :-

	Month	Year	Gross_Sales_Amount
►	September	2019	4496259.67
	October	2019	5135902.35
	November	2019	7522892.56
	December	2019	4830404.73
	January	2020	4740600.16
	February	2020	3996227.77
	March	2020	378770.97
	April	2020	395035.35
	May	2020	783813.42
	June	2020	1695216.60
	July	2020	2551159.16
	August	2020	2786648.26
	September	2020	12353509.79
	October	2020	13218636.20
	November	2020	20464999.10
	December	2020	12944659.65
	January	2021	12399392.98
	February	2021	10129735.57
	March	2021	12144061.25
	April	2021	7311999.95
	May	2021	12150225.01
	June	2021	9824521.01
	July	2021	12092346.32
	August	2021	7178707.59

Insights :-

This report analyze the monthly gross sales trend of Atliq Exclusive. It provides insights into seasonality, peak sales months, and performance fluctuations throughout the year.

For AtliqExclusive, November 2020 marked the highest sales, and March 2020 marked the lowest gross sales. It's very evident that the lower sales between March and August are because of COVID-19. However, it's a very good sign that the sales increased quickly after August and reached the highest level since the last two years in November.



Request 8 :-

In which quarter of 2020, got the maximum total_sold_quantity?

The final output contains these fields sorted by the total_sold_quantity,

- Quarter
- total_sold_quantity

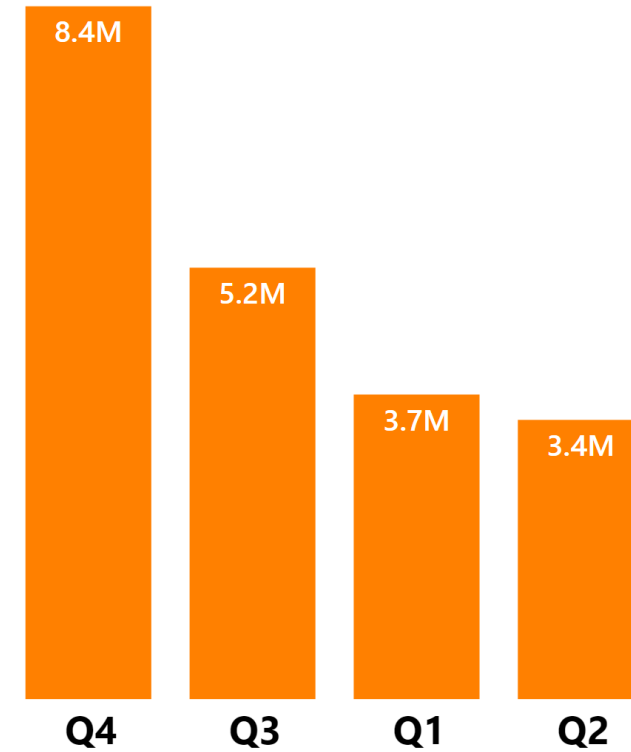
Output :-

	Quarter	total_sold_quantity
►	Q4	8425822
	Q3	5246770
	Q1	3704398
	Q2	3395899

Insights :-

This analysis determines which quarter recorded the highest total sold quantity in 2020. It helps identify peak demand periods and seasonal sales patterns.

This again complements the previous insight. That is the effect of COVID-19 on our sales. The sold quantity decreased to 3.4 million in quarter 2 of FY 2020, when COVID-19 was at its peak. But we started recovering very early despite the continuance of the pandemic. This early recovery during quarter 4 is probably because of the increased need for hardware like desktops and notebooks as majority of the students began or continued to do their coursework online during this time, and there was a huge demand for computer accessories during this period.



Request 9 :-

Which channel helped to bring more gross sales in the fiscal year 2021 and the percentage of contribution?

The final output contains these fields

- channel
- gross_sales_mln
- percentage

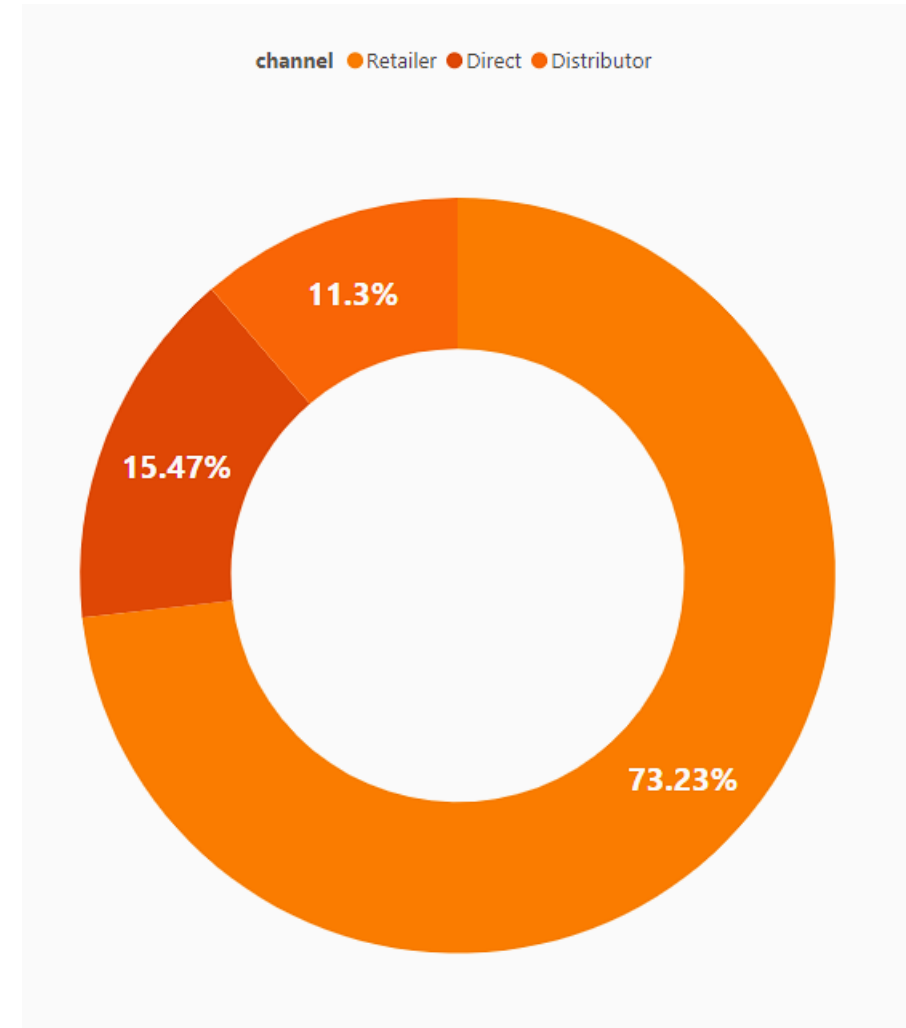
Output :-

	channel	gross_sales_mln	percentage
▶	Retailer	1219.08	73.23
	Direct	257.53	15.47
	Distributor	188.03	11.30

Insights :-

This report evaluates which sales channel contributed the highest revenue in FY2021 and calculates its percentage contribution to total sales.

The majority of our sales took place via retailers, which is 73% of the total sales. Only a very small percentage of our sales happened through direct and distributor channels.



Request 10 :-

Get the Top 3 products in each division that have a high total_sold_quantity in the fiscal_year 2021?

The final output contains these fields

- division
- product_code
- product
- total_sold_quantity
- rank_order

Output :-

	division	product_code	product	total_sold_quantity	rank_order
▶	N & S	A6720160103	AQ Pen Drive 2 IN 1	701373	1
	N & S	A6818160202	AQ Pen Drive DRC	688003	2
	N & S	A6819160203	AQ Pen Drive DRC	676245	3
	P & A	A2319150302	AQ Gamers Ms	428498	1
	P & A	A2520150501	AQ Maxima Ms	419865	2
	P & A	A2520150504	AQ Maxima Ms	419471	3
	PC	A4218110202	AQ Digit	17434	1
	PC	A4319110306	AQ Velocity	17280	2
	PC	A4218110208	AQ Digit	17275	3

Insights :-

This analysis identifies the top three best-performing products in each division based on total sold quantity in FY2021. It highlights key revenue-driving products within each division.

- The top 3 selling products in N&S were **pen drives**, which were around **7 lakh in quantity**.
- The top 3 selling products in P&A were **mouse**, which were around **4 lakh in quantity**.
- The top 3 selling products in PC were **personal laptops**, which were around **17000 in quantity**.

Top 3 Products by Division – FY 2021

Division	Total Sold Quantity
N & S	2065621
AQ Pen Drive 2 IN 1	
A6720160103	701373
AQ Pen Drive DRC	
A6818160202	688003
A6819160203	676245
P & A	1267834
AQ Gamers Ms	
A2319150302	428498
AQ Maxima Ms	
A2520150501	419865
A2520150504	419471
PC	51989
AQ Digit	
A4218110202	17434
A4218110208	17275
AQ Velocity	
A4319110306	17280
Total	3385444

THANK YOU



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